Hierarchical MARL for Distributed-Power × Electrical/Pneumatic Braking Division of Heavy-Haul Trains on Long Downhill Sections

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Abstract—Heavy-haul trains on sustained long downhill sections must hold speed within a narrow band using cyclic braking (electric + pneumatic), while keeping in-train coupler forces below a safety limit — a long-horizon, switched, safety-critical control problem. We show that the decomposition of this problem determines whether multi-agent reinforcement learning (MARL) can learn anything at all: prior "train-level" MARL treated each whole train as an agent and only nudged a shared speed-band threshold (a ±0.5 km/h offset = 1.25% of bandwidth), leaving a lever too small to learn from (a "nothing-to-learn" ceiling). We re-decompose at the locomotive level: each locomotive is an independent agent that directly modulates its own electric-braking depth and pneumatic-braking trigger — a continuous, physically meaningful lever. A rule engine (ED-first + inter-locomotive stagger + AB fallback) provides a safety-embedded inner layer; MAPPO adds per-locomotive $[\mathrm{ed}_{mod}, \mathrm{ab}_{mod}] \in [-1, 1]^2$ modulation. A semi-MDP decision interval splits the 15,000-step credit chain into 1,500 decisions, making long-horizon MAPPO converge. On the real Shuozhou–Huanghua profile (23.8 km, ≈ −10.9‰) with a 10,000-t, 1+1 distributed-power consist, we obtain a three-layer positive evidence chain with zero safety violations: single-locomotive rule (+40,956) → distributed-power rule division (+71,252; +74% reward, AB usage −89%) → RL division (+72,743; coupler peak −21% to 570 kN, final speed +11.2 km/h, ATP emergency events −97%). The learned policy is physically interpretable — halve ED depth and inject staggered minimum-notch AB to cut longitudinal impulse — reaching a region no tuning of the old threshold coordination could attain.

Index Terms—Heavy-haul train, distributed power, electro-pneumatic braking division, multi-agent reinforcement learning, cyclic braking, long downhill, coupler force.

I. Introduction

A 10,000-ton heavy-haul train on a long downhill section (Shuozhou–Huanghua line, 23.8 km, sustained up to −10.9‰) must regulate speed through cyclic braking: brake above an upper speed limit, release below a lower limit. The minute-scale brake–release cycles, combined with multi-body coupler dynamics and finite air-brake wave propagation (≈ 200 m/s across a ∼ 4 km train), generate in-train coupler forces that must stay below 1000 kN (normal) / 2250 kN (emergency) [9]. Electric braking (ED) is smooth, continuous, and available only on locomotives; pneumatic (air) braking (AB) is discrete (RELEASE/MIN/SERVICE/EMERGENCY) and is the dominant source of longitudinal impulse while simultaneously being slow to recover (pipe recharge). How to split braking effort between ED and AB across multiple locomotives — the distributed-power (DP) × electro-pneumatic division problem — is exactly where controller design matters most on long downhills [1], [2].

Our first attempt at MARL for this regime followed the literature's "train-level" decomposition [12]: one agent per train, outputting a small offset to a shared speed-band threshold. The result was a hard ceiling. The offset is ±0.5 km/h against a 40–80 km/h band — a 1.25% bandwidth lever — and across a 15,000-step run its reward gradient is statistically invisible. Adding global observations and consistency-shaped rewards recovered only ≈ 0.8% (∼ 343) reward. We diagnosed this as a structural problem, not a tuning problem: MARL has nothing to learn when its only control lever is too small to change the outcome.

The resolution is to re-decompose the action space at the locomotive level and give MARL a real physical lever:

- Electric-braking depth — each locomotive's $ED \in [0, 1]$ is continuous and directly scales the applied ED force (big lever, ±0.5 of absolute ED depth).
- Air-brake triggering — each locomotive may suppress or pre-trigger its own AB source along the consist (inter-locomotive stagger of pneumatic application), the primary knob for coupler impulse.

A rule engine provides the safety-embedded inner layer: ED-first, per-locomotive stagger, AB only as a fallback when ED availability is power-starved. MAPPO then learns how to modulate this baseline per locomotive. Because both ED depth and the AB on/off timing are physically meaningful, the RL policy can move coupler force, braking energy, and holding accuracy together.

A second, orthogonal obstacle is credit assignment over a very long horizon. The full 23.8 km section is a 15,000-step time domain; single-step-decision MAPPO over that horizon produced a flat, negative training reward (the RL policy ended pervasively dragging the train with air braking). We cure this with a semi-MDP decision interval: the RL policy decides once every N=10 steps, holds its action, and stores a window-summed transition —

compressing the 15,000-step credit chain to 1,500 decisions while the rule baseline still executes per step. This is the general technique on which the headline results depend.

Contributions. (C1) We identify and formalize the "nothing-to-learn" ceiling of train-level threshold coordination (a 1.25% bandwidth lever) and show that re-decomposition to the locomotive ED/AB level provides a physically meaningful MARL lever. (C2) We give a three-layer, like-for-like positive evidence chain on the real Shuozhou profile — single-locomotive → distributed-power rule division → distributed-power RL division — all with zero safety violations, isolating the architecture gain from the learning gain. (C3) We establish semi-MDP decision intervals as a practical cure for long-horizon (15,000-step) MARL credit assignment in switched train dynamics. (C4) The learned policy is physically interpretable — halve ED depth + staggered MIN-notch AB — and cuts coupler peak by 21%, raises final speed by 11.2 km/h, and reduces ATP-triggered emergency events by 97%.

II. Related Work

Distributed power × braking division. Tang et al. [1] model multi-locomotive trains with RL and explicitly include coupler force in the reward, establishing the DP setting we build on. Zhao et al. [2] give a refined air-brake model and report, for a 1+1 distributed-power consist, that synchronous multi-source braking reduces in-train impulse (their two-source ≈ 616 kN vs. single-source ≈ 1549 kN in our re-derivation), directly supporting the "ED/AB split + stagger" lever. Our contribution is to make this division learned per locomotive via MARL rather than hand-tuned.

RL for cyclic braking on long downhills. Zhang et al. [3], Liu et al. [4], and Wang et al. [5] all apply DQN-type RL to long-downhill automatic speed control — and all explicitly ignore coupler forces. We retain their cyclic-braking formulation but add coupler force as a first-class objective/constraint (1000 kN), which is precisely where the ED/AB division matters (AB is the impulse source). Zhang et al. [6] and Yang et al. [7] document that coupler force stays below 1000 kN on mild grades when braking is released simultaneously and pipe recharge is sufficient — the physics our dynamic multi-body model reproduces. Fan et al. [8] quantify how consist marshalling shapes longitudinal impulse in cyclic braking.

Why train-level MARL has "nothing to learn." Prior group-platoon MARL (e.g., the multi-train formulation in the earlier draft of this work) gave each train a cooperative threshold offset. The offset is ±0.5 km/h on a 40-km/h band → a 1.25% lever → near-zero reward gradient (our diagnostics). We report this as a structural caution for MARL action design: the lever must be able to change the outcome, or the gradient vanishes regardless of observation/architecture [11], [12].

Long-horizon credit assignment. PPO/MAPPO over tens of thousands of steps is known to degrade when the reward that matters arrives after many decisions. Our semi-MDP decision interval (action hold + window return) is a minimal, task-agnostic mitigation that preserves the underlying dynamics while shortening the credit chain; we include it as a reusable technique.

III. Problem Statement and System Model

Longitudinal dynamics per vehicle (NeTrainSim-derived multi-body integration): traction − electric braking − pneumatic braking − Davis resistance ± grade − coupler force. Air braking propagates along the consist at finite wave speed (≈ 200 m/s → seconds of delay across a ∼ 4 km train); electric braking is continuous and limited by power and adhesion. Coupler forces come from a mass–spring–damper with a gap dead-zone (0.01 m), reproducing the "coupler gap → coupler force" law [7] and Newton-pair cancellation.

Control objective: keep each locomotive's speed in $[v_{\min}, v_{\max}] = [40, 80]$ km/h, keep coupler force ≤ 1000 kN, over the full 23.8 km (15,000-step) horizon, using a 1+1 distributed-power consist (10,000 t). The difficulty is that the minute-scale cyclic rhythm and the band-limited coupler constraint act over far longer than a rolling window, and the AB actuator is discrete + history-dependent (recharge).

Why ED/AB splitting resists a small-lever policy. On this profile the rule-based cyclic braking already holds the band; the remaining degrees of freedom — exactly how deep each locomotive brakes with ED, and exactly when each triggers AB — are the ones that decide coupler impulse. A policy that cannot touch these (e.g., only shifts the band) is orthogonal to the objective it must minimize.

IV. Proposed Method: Two-Level Decompose MARL

Level 1 — rule engine (safety-embedded inner layer). A deterministic baseline get_action_decomposed runs per step: ED-first (both locomotives saturate ED up to the physically available share), per-locomotive stagger (offset each locomotive's brake phase to spread impulse), and AB fallback only when ED availability is insufficient. Critical to correct triggering is a physical ED-availability model:

$$F_{ED} = \min\big(P_{ED}/v,\ F_{\text{adhesion}}\big), \qquad F_{\text{req}} \propto m\,g\,\max|\text{grade}|. \tag{1}$$

On a steep section at speed the available ED share < 1, so the baseline genuinely falls back to AB; earlier simplifications (clip$(v/5)$) made the fallback unreachable and hid the mechanism.

Level 2 — MARL modulation (learned). MAPPO, one agent per locomotive (1+1 consist → 2 agents), observes per-locomotive state and outputs $[ed_{mod}, ab_{mod}] \in [-1, 1]^2$:

- ed_{mod} directly scales this locomotive's ED depth (±0.5 absolute) — the continuous physical lever;
- $ab_{mod} > +0.5$ suppresses this locomotive's AB source (only while $v < v_{\max} - 0.5$, so the safety fallback

is never cheaply switched off); $ab_{mod} < -0.5$ pre-triggers a MIN-notch AB — enabling staggered, anticipatory pneumatic braking.

Observation (11-dim per locomotive). Speed error w.r.t. target; margin to $v_{\max}$; margin to $v_{\min}$; the rule baseline's ED (what the engine "wants"); whether the rule baseline would brake with AB; local coupler force (normalized); AB-active fraction on the consist; AB release-remaining fraction (recharge state); brake-segment index; locomotive position (for grade inference); inter-train gap error (0 for single-train). Feeding the air-brake state lets the policy sense hysteresis rather than predict it.

Reward. Band holding (in-band +5, out-of-band −20 per train) + coupler force (mean & peak penalties, plus a super-limit term) + comfort + gap (multi-train extension) + an ED micro-cost (−0.02 per depth) to avoid gratuitous full-ED cruising. Notably, AB is not directly penalized: extensive smoke-testing showed any positive AB cost exactly cancels the coupler-force benefit of staggered MIN-notch AB (coupler force is the objective; AB is only a means).

Semi-MDP decision interval. With a per-step decision and a 15,000-step horizon, MAPPO's stage-3 training reward stayed flat at ≈ −291 k — the credit chain was too long. With decision interval N=10, the RL policy decides every 10 steps, holds its action, and stores one transition with a window-summed return; the rule baseline still executes every step. This turned a failed run into the converged results below.

V. Experiments

Protocol. Shuozhou–Huanghua main profile (23.8 km), 1+1 distributed power (10,000 t), ETCS distance-to-go curves on, deterministic evaluation (fixed seed) at 80 km/h over the full 15,000 steps, $n_{\text{seed}} = 3$ evaluation rollouts. MAPPO: lr $= 5 \times 10^{-5}$, entropy = 0.002, curriculum 1000:0, 3000:20, 15000:40, decision interval N=10, 60 episodes. The same environment + same rule baseline are used for M0 and the single-locomotive runs — the three rows of Table I differ only in the architecture/structure under test (like-for-like).

Isolated increments. (a) Architecture gain (M0 vs. single-loco): distributed-power ED-first + stagger + ED-availability-faithful AB fallback gives +74% reward (+30,278), +9.5 pp band rate, and −89% AB usage (4,030→461 steps) — distributed ED power suffices, so pneumatic effort almost vanishes, with zero super-limit. (b) Learning gain (M1 vs. M0): RL finds a physically interpretable re-allocation — halve ED depth (mean ED 0.67 → 0.35) and inject staggered MIN-notch AB (461→4,357 steps) — trading slightly more, but individually gentle and phased, pneumatic applications to shrink longitudinal impulse: coupler peak −21% (720→570 kN), final speed +11.2 km/h, ATP emergency events −97% (43.2 M→1.44 M). The smoother ED profile plus phased AB is exactly the region a small-lever threshold policy can never reach.

Why this is the decisive comparison. The three rows share the environment, the safety layer, and the rule baseline; only whether (i) power is distributed and (ii) ED/AB are RL-modulated per locomotive changes. Every layer adds a positive, measured increment with zero violations — decomposing the source of the gain (architecture vs. learning) rather than reporting a single black-box number. The same simulator, given a real lever, transitions from a flat −291 k training reward to +72.7 k rollout reward.

Ablation — lever isolation (which lever carries the gain). We reran full training (seed0, same protocol) with one modulation masked during training (mask_lever), isolating the causal contribution of each lever (best-checkpoint evaluation; all zero violations, band ≈ 100%): Table II.

The two levers serve distinct objectives. The ED lever is the coupler-safety lever: ED-only reaches the lowest coupler peak (528 kN, 27% below M0) with zero AB and 100% band, so the coupler reduction of M1 vs. M0 comes chiefly from ED-depth shaping, not from AB. The AB lever is the throughput lever: AB-only rides the speed-band ceiling (final 71.0 km/h, heavy phased AB ≈5,960 steps, near-zero ED) and scores the highest scalar reward (+74,634), at the cost of higher coupler force (702 kN). M1 (both levers) is the compromise — 578 kN, +73,733, final 53.5 km/h — jointly minimizing coupler force and holding throughput. Honest caveat: only-AB outperforms M1 on the raw scalar reward, i.e., the current reward shaping rewards high-speed band-riding air-braking; we therefore do not claim M1 is the scalar-reward optimum, but the best coupler–throughput balance among safe (zero-violation) policies. Reshaping to penalize heavy AB (impulse/energy, not a linear cost) is a stated follow-up.

Generalization / recentring. Because the policy acts on per-locomotive ED/AB (not on a train-wide band), it transfers structurally to longer sections and larger numbers of distributed-power locomotives without retuning the band; ablations over $N \in \{5, 10, 20\}$ and AB-cost $\in \{0, 0.05\}$ (planned) will establish the margins of the semi-MDP interval and the reward shape.

VI. Conclusion and Limitations

We argued and demonstrated that for long-downhill cyclic-braking control, the decomposition of the MARL action space decides whether there is anything to learn. Train-level threshold coordination gives a 1.25% bandwidth lever and a near-zero gradient ("nothing to learn"); locomotive-level ED/AB division gives continuous, physically meaningful levers. Combining a safety-embedded rule engine with per-locomotive MAPPO modulation and a semi-MDP decision interval yields a three-layer, zero-violation positive evidence chain on the real Shuozhou

TABLE I
Main result — three-layer evidence chain (15,000 steps, all zero safety violations).

Method	reward	coupler peak (kN)	viol	band rate	AB steps	mean ED	final speed (km/h)	ATP events
decompose-M1 (RL division, N=10, ep60)	+72,743	570	0	99.72%	4,357	0.35	52.9	1.44 M
decompose-M0 (distributed-power rule division)	+71,252	720	0	99.68%	461	0.67	41.7	43.2 M
single-locomotive (old 1+0 rule)	+40,956	456	0	90.20%	4,030	—	44.9	467 M

TABLE II
Lever-isolation ablation (seed0 best checkpoint).

Method	reward	coupler (kN)	AB steps	mean ED	final (km/h)
M1 both levers (seed0 best)	+73,733	578	4,807	0.34	53.5
only-ED lever (mask ab)	+71,932	528	0	0.70	45.1
only-AB lever (mask ed)	+74,634	702	5,960	0.05	51.0
M0 rule (no levers)	+71,252	720	461	0.67	41.7

profile (RL: +2.1% reward over rule division; coupler −21%; final speed +11.2 km/h; ATP events −97%).

Limitations. (1) The learned policy is trained in a high-fidelity simulator and inherits its fidelity (a sim-to-real gap shared in kind with model-driven MPC). (2) We provide no formal recursive-stability guarantee; deployment safety would require formal verification / ATP integration. (3) AB is not directly penalized in the reward — the division is only as safe as the coupler-force weighting, and the AB-trigger margin ($v_{\max} - 0.5$) is a hand-set constant. (4) Long-horizon training remains expensive and seed variance is nontrivial; the semi-MDP interval N is a hyperparameter worth ablating. (5) Inter-train gap adaptation (platoon level) is a stated extension; this paper fixes a single train with 1+1 distributed power. (6) The spring-buffer (draft-gear) model introduces a few calibrated constants: the MT-2 static/dynamic friction ratio $\mu_s/\mu_d = 1.5$ and transition velocity $v_{\text{er}} = 0.2$ m/s have a literature-anchored structure (Yang Liangliang 2017, Eq. 3) but point values from impact tests; the buffer is unified to a single MT-2 friction type (including the locomotive–freight connection), avoiding an elastic-paste sub-model whose five shape constants had no published fits; our Shuozhou-profile sweep shows the coupler-force peaks shift $\leq$1% within reasonable uncertainty of these values (v_{er} 0.05–0.5 $\rightarrow$ +1%; μ_s/μ_d 1.2–1.5 $\rightarrow$ +1.2%), so they do not dominate the headline coupler-force result (though μ_s/μ_d shows a threshold nonlinearity near 1.8). Also, the release-time constant $\tau = 6$ s models only the initial exhaust transient (exponential decay); full air-brake recharge takes 175–512 s (Zhao 2022), which we abstract as a history-dependent recharge effect rather than simulate explicitly.

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